

Proofs Involving Disjunctions

NETSOL

3.5
Pg 150.

Date _____

MTWTFSS

$$1) \forall x (x \in A \cap (B \cup C) \rightarrow x \in (A \cap B) \cup C)$$

$$\text{suppose } x \in A \cap (B \cup C)$$

$$x \in (A \cap B) \cup C$$

$$x \in A \text{ --- (1)}$$

$$(x \in A \wedge x \in B) \vee x \in C$$

$$x \in B \vee x \in C$$

$$\downarrow \quad \vee \quad \downarrow$$

$$x \in B \vee x \in C \text{ --- (2)}$$

Case 1) suppose $x \in B$ then using (1), we proved goal

Case 2) suppose $x \in C$ then goal is T.

$$2) \forall x [x \in (A \cup B) \cap C \rightarrow x \in A \cup (B \cap C)]$$

$$\text{suppose } (x \in A \vee x \in B) \wedge x \in C$$

$$x \in A \vee (x \in B \wedge x \in C)$$

$$x \in C$$

$$x \in A \vee x \in B$$

Case 1) suppose $x \in A$ Proved

$$\text{Case 2) } x \in B \rightarrow x \in B \wedge x \in C$$

$$3) A \cap (A \cup B) = A \cap B$$

$$\forall x [x \in A \cap (A \cup B) \leftrightarrow x \in A \cap B]$$

$$\rightarrow \text{suppose } x \in A \cap (A \cup B)$$

$$x \in A \cap B$$

$$x \in A \wedge \neg(x \in A \wedge x \notin B)$$

$$x \in A \wedge x \in B$$

$$x \in A$$

$$x \notin A \vee x \in B$$

Since $x \notin A \vee x \in B$ but we know that

$x \in A$ so $x \in B$ Proved. $(\because p \vee q, \neg p \therefore q)$

$$\forall x [x \in A \cap B \longrightarrow x \in A \cap (A \cap B)]$$

Suppose $x \in A \cap x \in B$

$$\begin{aligned} & \longrightarrow x \in A \\ & \quad x \in A \vee x \in B \end{aligned}$$

$$4) \forall x [x \in A \cap (B \cap C) \iff x \in (A \cap B) \cup (A \cap C)]$$

→ Suppose $x \in A \cap \neg(x \in B \cap x \in C)$

$$x \in A$$

$$x \in A \cap B \vee x \in A \cap C$$

$$x \notin B \vee x \in C$$

$$(x \in A \cap x \notin B) \vee (x \in A \cap x \in C)$$

Earlyly both cases follow.

← Suppose $x \in (A \cap B) \vee x \in (A \cap C)$

$$(x \in A \cap x \in B) \vee (x \in A \cap x \in C)$$

$$x \in A \cap (x \in B \vee x \in C)$$

$$= (x \in A \cap (x \in B \vee x \in C)) \text{ --- Associative}$$

Case 1) → both follow as $x \in A$ & from $x \in B \vee x \in C$

one of it is T as $x \in B$.

Case 2 also follows.

$$5) A \cap C \subseteq B \cap C$$

$$A \cup C \subseteq B \cup C$$

$$A \subseteq B$$

$$\forall x (x \in A \rightarrow x \in B)$$

Let x be arbitrary

$$\forall x [(x \in A \cap x \in C) \rightarrow (x \in B \cap x \in C)] \text{ --- ①}$$

$$\forall y [(y \in A \cup y \in C) \rightarrow (y \in B \cup y \in C)] \text{ --- ②}$$

Suppose $x \in A$

$$x \in B$$

$x \in B \vee x \in C$ — Universal inst ②

Case 1) $x \in B$ Proved.

Case 2) $x \in C$. Then as $x \in A, x \in B \wedge x \in C \therefore$ universal inst ①. $\therefore x \in B$. Proved.

b) Suppose A, B be arbitrary sets

Suppose $A \supset B \subseteq A$

$B \subseteq A$

$\forall x [x \in A \supset B \rightarrow x \in A]$

$\forall y [y \in B \rightarrow y \in A]$

$\forall x [x \in (A \setminus B) \cup (B \setminus A) \rightarrow x \in A]$

$\forall x [x \in (A \setminus B) \vee x \in (B \setminus A) \rightarrow x \in A] \text{ --- ①}$

But y is arbitrary
contradiction

Suppose $y \in B$ & $y \notin A$

so $y \in B \setminus A \text{ --- ②}$

From ① & ② $\rightarrow y \in A \therefore$ Universal Inst

which is a contradiction

7) ~~$\forall x [x \in A \vee C]$~~

$A \cup C \subseteq B \cup C \iff A \setminus C \subseteq B \setminus C$

$\rightarrow$ Suppose $A \cup C \subseteq B \cup C$

$\forall x [x \in A \cup C \rightarrow x \in B \cup C]$

$\forall x [(x \in A \vee x \in C) \rightarrow (x \in B \vee x \in C)]$

$\forall y [y \in A \setminus C \rightarrow y \in B \setminus C]$

Let y be arbitrary

Suppose $y \in A \setminus C$

$y \in B \setminus C$

$\rightarrow y \in A$

$y \in B \wedge y \notin C$

$y \notin C$

As $y \in A$ so $y \in B \vee y \in C$. We know that $y \notin C$, so $y \in B$

so $y \in B$ & $y \notin C$. Proved

← Suppose $A \setminus C \subseteq B \setminus C$

$A \cup C \subseteq B \cup C$

$\forall x [x \in A \setminus C \rightarrow x \in B \setminus C]$

$\forall y [y \in A \cup C \rightarrow y \in B \cup C]$

$\forall x [(x \in A \wedge x \notin C) \rightarrow (x \in B \wedge x \notin C)]$ let y be arbitrary

Suppose $y \in A \cup C$

$y \in B \cup C$

$y \in A \vee y \in C$

$y \in B \vee y \in C$

Direct Proof is difficult. Contraposition also. We will prove by contradiction.

Suppose $A \setminus C \not\subseteq B \setminus C$

$\exists x [(x \in A \wedge x \notin C) \rightarrow x \notin B \wedge x \notin C]$ (a)

Suppose $A \cup C \not\subseteq B \cup C$

So $\exists y [y \in A \cup C \wedge y \notin B \cup C]$

$y_0 \in A \vee y_0 \in C$ - ① :- existential

$y_0 \notin B \wedge y_0 \notin C$ - ② inst

Since $y_0 \notin C$, so $y_0 \in A$ from ①, ②.

So $y_0 \in A$ - ③

$y_0 \notin B$ - ④

$y_0 \notin C$

So $y_0 \in A \setminus C$ - ③④

So $y_0 \in B \setminus C$ - ⑤ (inst)

So $y_0 \in B \wedge y_0 \notin C$

But $y_0 \notin B$ is a contradiction

Actually here direct proof is given. Our goal is to prove $A \cup C \subseteq B \cup C$.
If $y \in C$, then there is nothing more to prove. Now suppose $y \notin C$, from given $\rightarrow$ we can conclude $y \in A$, so $y \in A \setminus C$. So $y \in B \setminus C \rightarrow$ so $y \in B$.

$$8) \forall x [x \in P(A) \cup P(B) \rightarrow x \in P(A \cup B)]$$

$$\text{Suppose } x \in P(A) \cup P(B)$$

$$x \in P(A \cup B)$$

$$x \in P(A) \vee x \in P(B)$$

$$x \subseteq (A \cup B)$$

$$x \subseteq A \wedge x \subseteq B \quad \text{--- (1)}$$

$$\forall y [y \in x \rightarrow y \in A \cup B]$$

why is arbitrary

$$y \in A \vee y \in B$$

Suppose $y \in x$

From (1), we have 2 cases

Case 1) $x \subseteq A$, since $y \in x$, so $y \in A$

Case 2) $x \subseteq B$, as $y \in x$, so $y \in B$

$$a) \text{ Suppose } P(A) \cup P(B) = P(A \cup B)$$

$$\forall x [x \in P(A) \vee x \in P(B) \rightarrow x \in P(A \cup B)] \quad \text{--- (1)}$$

$$A \subseteq B \vee B \subseteq A$$

$$\forall x [x \in P(A \cup B) \rightarrow x \in P(A) \vee x \in P(B)] \quad \text{--- (2)}$$

$\therefore$ Universal quantifier Distributes over conjunction
Let x be arbitrary

$$\text{--- (1)} \rightarrow x \subseteq A \vee x \subseteq B \rightarrow x \subseteq (A \cup B) \quad \text{--- (3)}$$

$$\text{--- (2)} \quad x \subseteq (A \cup B) \rightarrow x \subseteq A \vee x \subseteq B \quad \text{--- (4)}$$

We have to use the fact that every set is subset of itself.

Since $A \cup B \subseteq A \cup B$

$$\text{From (4)} \rightarrow A \cup B \subseteq A \wedge A \cup B \subseteq B$$

Case 1) $A \cup B \subseteq A$ - Suppose $x \in B$, then $x \in A$, so $B \subseteq A$.

Case 2) $A \cup B \subseteq B$ - Suppose $y \in A$, then $y \in B$, so $A \subseteq B$

$$10) \begin{array}{l} x \in \mathbb{R} \\ y \in \mathbb{R} \\ x \neq 0 \end{array}$$

$$y + \frac{1}{x} = 1 + \frac{y}{x} \text{ iff } x=1 \vee y=1$$

$$\rightarrow \text{Suppose } y + \frac{1}{x} = 1 + \frac{y}{x}$$

$$x=1 \vee y=1$$

If $x=1$, then there is nothing more to prove

And if $x \neq 1$, then

$$y + \frac{1}{x} = 1 + \frac{y}{x}$$

$$\Rightarrow \frac{xy+1}{x} = \frac{x+y}{x}$$

Since $x \neq 0$, so we divide by x

$$= xy + 1 = x + y$$

$$= x + y - xy = 1$$

$$= x + y(1-x) = 1$$

$$y = \frac{1-x}{1-x} \rightarrow y=1$$

$\leftarrow$ Suppose $x=1 \vee y=1$

$$y + \frac{1}{x} = 1 + \frac{y}{x}$$

Case 1) $x=1$ then $y + \frac{1}{1} = 1 + \frac{y}{1} \Rightarrow y+1 = y+1$ True

Case 2) $y=1$ then $1 + \frac{1}{x} = 1 + \frac{1}{x}$ True

QED

11) $|x-3| > 3$ — ①

$x^2 > 6x$

Case 1 - Suppose $x-3 \geq 0$

then $|x-3| = x-3$ — ②

putting ② in ①

$x-3 > 3 \rightarrow x > 6$

Multiplying by $x \rightarrow x^2 > 6x$

Case 2) Suppose $x-3 < 0$

then $|x-3| = 3-x$

so $3-x > 3 \therefore ①$

taking square $3^2 + x^2 - 6x > 3^2$

$x^2 - 6x > 0 \rightarrow x^2 > 6x$

12) $|2x-6| > x$ iff $|x-4| > 2$

$\rightarrow$ Suppose $|2x-6| > x$ ①

$|x-4| > 2$

Case ① if $2x-6 \geq 0$, then $|2x-6| = 2x-6$

if $x-4 \geq 0$ then

so ① $\rightarrow 2x-6 > x \rightarrow x-6 > 0$

$x-4 > 2$

$\rightarrow x-4-2 > 0$

if $x-4 < 0$ then

$\rightarrow x-4 > 2$

$4x > 2$

Case ② Suppose $2x-6 < 0$

so $x-4 > 2 \vee 4-x > 2$

then $|2x-6| = 6-2x$

so $6-2x > x$ then $6 > 3x \rightarrow x < 2$

$\rightarrow -x > -2$ Adding 4 $\rightarrow 4-x > 2$

← Suppose $|x-4| > 2 \rightarrow$ (1)

$|2x-6| > x$

(1) if $x-4 > 0$ then $|x-4| = x-4$

$2x-6 > x$ or $6-2x > x$

so $x-4 > 2 \rightarrow x > 6$

$\therefore$ By def of mod

$\rightarrow$ Add x , $2x > 6+x$

so $2x-6 > x$

Case (2) if $x-4 < 0$ then $|x-4| = 4-x$

so $4-x > 2 \rightarrow x < 2 \rightarrow 3x < 6$

so ~~subtract~~ subtract $2x$, we get $x < 6-2x$

or $6-2x > x$

As some x can be > 0
or < 0 , these cases
are exhaustive

We can also prove
that these 2 cases lead
to +ve & negative mods
in the goal but
we already have 2 cases mod
in our goal so that's fine

13) (a) $a \in \mathbb{R}$

$\rightarrow |a| \leq b$ iff $-b \leq a \leq b$

Case (1) if $a \geq 0$ then $|a| = a$
so $a \leq b$

wrong. trying case 1 should
imply both $a \geq 0$ and $a \leq 0$
since $a \geq 0$ and $a \leq 0$
so $b \geq 0$, so $-b \leq 0$
so $-b \leq 0 \leq a \leq b$
so $-b \leq a \leq b$

Case (2) if $a < 0$ then $|a| = -a$
so $-a \leq b$ or $-b \leq a$

Combining these, we have $-b \leq a \leq b$ similar for case 2.

← Suppose $-b \leq a \leq b$

so $-b \leq a$ & $a \leq b$

$-b \leq a$

so $-a \leq b$

Case 2: since $a < 0$,
 $-a \leq b$ so $-b \leq a$ (also)
 $-a \geq 0$ so $-b \leq a < 0 < -a \leq b$
so $-b \leq a \leq b$

$|a| \leq b$

$a \leq b$

$a \leq b$ when $a \geq 0$ or $-a$ when $a < 0$

Proved.

(b) $x \in \mathbb{R}$

Case 1) When $x \geq 0$

then as $x \geq 0 \rightarrow -x \leq 0$

so $x \geq 0 \geq -x$

so $x \geq -x$ or $-x \leq x$ — (1)

Adding as $x \geq 0$

subtracting x , we get

$$0 \geq -x$$

dividing by

Clearly every number $x = x$

so $x \leq x$ — (2)

Looking (1) & (2) $-x \leq x \leq x$ and x is positive

so $-|x| \leq x \leq |x| = -x \leq x \leq x$ when $x \geq 0$.

Case 2) $x < 0$

Since $|x| \leq |x|$ by part (a) of Q13. It follows that $-|x| \leq x \leq |x|$

(c) $x, y \in \mathbb{R}$

$$|x+y| \leq |x|+|y|$$

We will use parts (a) and (b)

As $|x| \leq y$ iff $-y \leq x \leq y$ for every $x, y \in \mathbb{R}$

so $|x+y| \leq |x|+|y|$ iff $-|x|-|y| \leq x+y \leq |x|+|y|$

so $x+y \leq |x|+|y|$ — (1)

As $x \leq |x|$ for every $x \in \mathbb{R}$

so $x+y \leq |x+y|$ — (2)

subtracting (1) & (2) $\rightarrow 0 \leq |x|+|y| - (x+y) \rightarrow |x+y| \leq |x|+|y|$

(d) $x, y \in \mathbb{R}$

$$|x+y| \geq |x| - |y|$$

given $|x| = |(x+y) + (-y)|$

Applying triangle inequality

$$|x| = |(x+y) + (-y)| \leq |x+y| + |-y|$$

since $|-y| = |y|$ for every $y \in \mathbb{R}$

$$\text{so } |x| \leq |x+y| + |y|$$

$$\text{so } |x+y| \geq |x| - |y|$$

14) $n \in \mathbb{Z}$

$n^2 + n$ is even

Case 1 - Suppose n is even

$$\exists n \in \mathbb{Z} (n^2 + n = 2n)$$

then $n = 2k, k \in \mathbb{Z}$

$$\downarrow$$

$$n(n+1)$$

$$\text{so } n^2 + n = n(n+1)$$

$$= 2k(2k+1)$$

$$= 4k^2 + 2k = 2(2k^2 + k) \quad (\because \text{let } n = 2k^2 + 2k)$$

as $2k^2 + k \in \mathbb{Z}$, so $n^2 + n$ is even.

Case 2 - Suppose now n is odd.

then $n = 2k+1, k \in \mathbb{Z}$

$$\begin{aligned} n(n+1) &= (2k+1)(2k+2) \\ &= 4k^2 + 4k + 2k + 2 \\ &= 2(k^2 + 2k + k + 1) \end{aligned}$$

Since $k^2 + 2k + k + 1 \in \mathbb{Z}$, so $n(n+1)$ is even.

Since n can be either even or odd, so these cases are exhaustive.

15) $x \in \mathbb{Z}$

($x^4 \div 8$ has remainder 0) $\vee$ (1)

Case 1) x is even then

$$\exists k (x = 2k)$$

$$x^4 = (2k)^4 = 16k^4. 16k^4 \text{ is divisible by } 8, \text{ so remainder is } 0$$

Case 2) x is odd. Then $\exists k (x = 2k+1)$

$$x^4 = (2k+1)^4 = (2k)^4 + 4(2k)^3 + 6(2k)^2 + 4(2k) + 1$$

$$= 16k^4 + 32k^3 + 24k^2 + 8k + 1$$

$$= 8(2k^4 + 4k^3 + 3k^2 + k) + 1$$

so when x is odd, x^4 leaves a remainder of 1 when divided by 8

$$\therefore (a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} a^{n-n} b^n$$

16) (a) $F \neq \emptyset, G \neq \emptyset$

$$B \cup (U \cap F) = U \cap (F \cup \{B\})$$

$$\Rightarrow \text{Suppose } x \in B \cup (U \cap F) \quad x \in U \cap (F \cup \{B\})$$

$$U \cap (F \cup G) = (U \cap F) \cup (U \cap G)$$

$$\Rightarrow \text{Suppose } x \in U \cap (F \cup G)$$

$$x \in (U \cap F) \cup (U \cap G)$$

$$\exists A (A \in F \cup G \wedge x \in A)$$

$$x \in U \cap F \vee x \in U \cap G$$

$$\exists A [(A \in F \vee A \in G) \wedge x \in A]$$

$$\exists B (B \in F \wedge x \in B) \vee \exists C (C \in G \wedge x \in C)$$

$$A \in F \vee A \in G$$

$$x \in A$$

Case 1) Suppose $A \in F$. Also $x \in A$. So let $B = A$.

Case 2) Suppose $A \in G$. Also $x \in A$. So let $C = A$.

$$\leftarrow x \in (U \cap F) \cup (V \cap G)$$

$$x \in U \cap (F \cup G)$$

$$\exists B (D \in A \wedge x \in B) \vee \exists C (C \in G \wedge x \in C)$$

$$\exists A [(A \in F \vee A \in G) \wedge x \in A]$$

Case 1) Suppose $D \in F, x \in B_0$. So let $A = B_0$

Case 2) Suppose $C \in G, x \in C_0$. So let $A = C_0$

$$(b) \cap (F \cup G) = (\cap F) \cap (\cap G)$$

$$\rightarrow \text{Suppose } x \in \cap (F \cup G)$$

$$x \in \cap F \wedge x \in \cap G$$

$$\cancel{x \in F} \wedge \cancel{x \in G}$$

$$\forall A (A \in F \rightarrow x \in A) \wedge \forall B (B \in G \rightarrow x \in B)$$

$$\forall A (A \in (F \cup G) \rightarrow x \in A)$$

$$\forall A (A \in F \rightarrow x \in A)$$

$$\forall A [(A \in F \vee A \in G) \rightarrow x \in A]$$

$$\forall B (B \in G \rightarrow x \in B)$$

$$\text{Suppose } A \in F$$

$$x \in A$$

$$\text{Suppose } B \in G$$

$$\text{Universal instantiation } x \in B$$

$$\leftarrow \text{Suppose } x \in \cap F \cap \cap G$$

$$x \in \cap (F \cup G)$$

$$\forall A (A \in F \rightarrow x \in A) \text{ --- (1)}$$

$$\forall A [(A \in F \vee A \in G) \rightarrow x \in A]$$

$$\forall B (B \in G \rightarrow x \in B) \text{ --- (2)}$$

$$x \in A$$

$$\text{Suppose } A \in F \vee A \in G$$

$$\text{Case 1) Suppose } A \in F \rightarrow x \in A \text{ (1)}$$

$$\text{Case 2) Suppose } A \in G \rightarrow x \in A \text{ (2)}$$

$$(17) B \cup (U \cap F) = U \cap (F \cup \{B\})$$

$$\rightarrow x \in B \vee \exists A (A \in F \wedge x \in A)$$

$$\exists A' \in (F \cup \{B\}) \wedge x \in A'$$

$$\exists A [(A \in F \vee A \in \{B\}) \wedge x \in A]$$

Case 1) Suppose $x \in B$. It's a fact that $B \in \{B\}$ so we found a set such that $B: \{B\} \neq x \in B$

Case 2) Suppose $A \in F \wedge x \in A \rightarrow$ what follows.

$$\begin{aligned} \leftarrow x \in \bigcup (F \cup \{B\}) & \quad x \in B \cup \bigcup (F) \\ \exists A [(A \in F \vee A \in \{B\}) \wedge x \in A] & \quad x \in B \vee \exists A (A \in F \wedge x \in A) \\ \exists A [(A \in F \wedge x \in A) \vee (A \in \{B\} \wedge x \in A)] & \end{aligned}$$

3. Associative law

$$\begin{aligned} \exists A (A \in F \wedge x \in A) & \text{--- (1) : existential quantifier distributes over disjunctive} \\ \exists A (A \in \{B\} \wedge x \in A) & \text{--- (2)} \end{aligned}$$

Case 1) suppose (1) $A \in F \cdot x \in A$

Case 2) " (2) $A \in \{B\} \cdot$ so $x \in B$..
In fact.

$$(b) B \cup (\cap F) = \cap_{A \in F} (B \cup A)$$

$$\begin{aligned} \rightarrow x \in B \vee x \in \cap F & \quad x \in \cap_{A \in F} (B \cup A) \\ x \in B \vee \forall A (A \in F \rightarrow x \in A) & \text{--- (1)} \quad \forall A \in F (x \in B \cup A) \\ & \quad \forall A [A \in F \rightarrow (x \in B \vee x \in A)] \\ & \quad x \in B \vee x \in A \end{aligned}$$

Suppose $A \in F$ (2).

Case 1) $x \in B$. Proved

Case 2) $A \in F$. From (1) $\rightarrow x \in A$.

Suppose $\forall A (A \in F \rightarrow x \in A)$.

$$\begin{aligned} \leftarrow \text{suppose } x \in \cap_{A \in F} (B \cup A) & \quad x \in B \cup (\cap F) \\ \forall A [A \in F \rightarrow (x \in B \vee x \in A)] & \text{--- (1)} \quad x \in B \vee \forall A [A \in F \rightarrow x \in A] \end{aligned}$$

Suppose $A \in F$

$x \in B \vee x \in A$

From ①: $x \in A \vee x \in B$

$$(c) B \cap (U \cap F) = U \cap F \cap (B \cap A)$$

→ Suppose $x \in B \cap x \in U \cap F$

$x \in U \cap F \cap (B \cap A)$

$x \in B$ — ①

$\exists A \in F (x \in B \cap A)$

② $\exists A (A \in F \wedge x \in A)$

$\exists A [A \in F \wedge (x \in B \cap x \in A)]$

$A_0 \in F \wedge x \in A_0$ — ②

① and ② trail follow.

← $\exists A [A \in F \wedge x \in B \wedge x \in A]$

$x \in B \wedge \exists A [A \in F \wedge x \in A]$

$A_0 \in F, x \in B, x \in A_0$ →

$$(d) B \cap (A \cap F) = A \cap (F \cup \{B\})$$

→ Suppose $x \in B \cap x \in A \cap F$

$x \in A \cap (F \cup \{B\})$

$\forall A (A \in F \rightarrow x \in A)$ — ①

$\forall A [x \in F \cup \{B\} \rightarrow x \in A]$

$\forall C [(x \in F \vee x \in \{B\}) \rightarrow x \in C]$

Let C be arbitrary

Suppose $x \in F \vee x \in \{B\}$

$x \in C$

Suppose $x \in F$

From ① $\rightarrow x \in C$

←

Suppose $x \in A \cap (F \cup \{B\})$

$x \in B \cap (A \cap F)$

$\forall A [(x \in F \vee x \in \{B\}) \rightarrow x \in A]$ — ①

$x \in B \wedge \forall A (A \in F \rightarrow x \in A)$

Let A be arbitrary. Suppose $A \in F$.

$x \in B \wedge x \in A$.

$\therefore$ For questions of this type. Remember that

$$(p \vee q) \rightarrow r \\ = (p \rightarrow r) \wedge (q \rightarrow r).$$

Since $A \in F$, from ① $x \in A$

& since $B \in \{B\}$. From ① $\rightarrow x \in B$.

$$18) F \neq \emptyset, G \neq \emptyset, H \neq \emptyset$$

$$\forall A \in F \forall B \in G (A \cup B \in H).$$

$$\cap H \subseteq (\cap F) \cup (\cap G)$$

Let $A \in F$ & $B \in G$ be arbitrary

supp $A \cup B \in H$ ①

$$\forall x [x \in \cap H \rightarrow x \in (\cap F) \cup (\cap G)]$$

Suppose $x \in \cap H$

$$x \in (\cap F) \cup (\cap G)$$

$$\forall C (x \in C \rightarrow x \in C) \text{ --- ②}$$

$$x \in \cap F \vee x \in \cap G$$

As $A \cup B \in H$, so $x \in A \cup B$
 $\therefore$ ① & ②

$$\forall A (A \in F \rightarrow x \in A) \vee \forall B (B \in G \rightarrow x \in B)$$

$$x \in A \vee x \in B$$

$$x \in A \vee x \in B.$$

$\therefore$ As A & B are arbitrary

& $A \in F$ & $B \in G$

Both cases implies both goals.

$$19) A \text{ & } B \text{ are sets}$$

$$\forall x [x \in A \wedge B \subseteq \rightarrow (x \in A \rightarrow x \in B)]$$

Let x be arbitrary

$$\rightarrow \text{supp } x \in A \wedge B$$

$$x \in A \rightarrow x \in B$$

$$x \in (A \cap B) \cup (B \setminus A)$$

$$1) x \in A \rightarrow x \in B$$

$$x \in A \cap B \vee x \in B \setminus A$$

$$2) x \in B \rightarrow x \in A$$

$$(x \in A \wedge x \in B) \vee (x \in B \wedge x \in A)$$

$$x \in B$$

As $x \in A$ & $x \notin B$ goal fails

suppose $x \notin B$

$x \in A$

Case 2) Suppose $x \notin B \wedge x \in A$ so $x \in A$.

← Suppose $x \in A \rightarrow x \notin B$

$x \in A \rightarrow x \notin B$

$x \notin B \rightarrow x \in A$

$x \in A^c \cap B$

$(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)$

Suppose $x \in A$ then $x \notin B$

Suppose $x \notin B$ then $x \in A$.

∴ arbitrary choice

20) $A \cap C = B \cap C$

$\forall y [y \in A \cap C \rightarrow y \in B \cap C]$

$\forall y [(y \in A \wedge y \in C) \rightarrow (y \in B \wedge y \in C)]$

$A^c \cap B$ & C are disjoint

$\forall x [x \in A^c \cap B \rightarrow x \notin C]$

but x is arbitrary

suppose $x \in A^c \cap B$

$(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)$

$x \notin C$

$\forall y [(y \in A \wedge y \in C) \rightarrow (y \in B \wedge y \in C)]$ ①

$\forall y [(y \in B \wedge y \in C) \rightarrow (y \in A \wedge y \in C)]$ ②

suppose $x \in C$

contradiction

Case 1) $x \in A \wedge x \notin B$. Since $x \in C$. From ① $x \in B \wedge x \in C$ but $x \notin B$ is a contradiction

Case 2) $x \in B \wedge x \notin A$. From ② since $x \in C$, so $x \in A \wedge x \in C$. which is a contradiction since $x \notin A$.

← $A^c \cap B$ & C are disjoint

$A \cap C = B \cap C$

$\forall x [x \in A^c \cap B \rightarrow x \notin C]$

$\forall y [(y \in A \wedge y \in C) \rightarrow (y \in B \wedge y \in C)]$

$\forall x [(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A) \rightarrow x \notin C]$

$\forall y [(y \in B \wedge y \in C) \rightarrow (y \in A \wedge y \in C)]$

$$\forall x [x \in C \rightarrow x \in A \cap B]$$

$$\forall x [x \in C \rightarrow (x \in A \vee x \in B) \wedge (x \in B \vee x \in A)]$$

Suppose $y \in A$ & $y \in C$ is arbitrary

$$\text{Since } y \in C \text{ so } (y \in A \vee y \in B) \wedge (y \in B \vee y \in A)$$

$$\text{Since } y \in A \text{ so } y \in B$$

$$\text{so } y \in A \cap C \rightarrow y \in B \cap C$$

$$\text{or } A \cap C \subseteq B \cap C \quad \textcircled{1}$$

$$\therefore p \vee q$$

$$\neg p \therefore q$$

The other is contradictory. If $y \in A$ but $y \notin B$ then $y \notin A \cap B$ but $y \in C$ so $y \in A \cap C$ but $y \notin B \cap C$ so $A \cap C \not\subseteq B \cap C$ so this will not be disjoint.

$$\text{Now suppose } y \in B \cap C \text{ then } y \in C \text{ so } (y \in A \vee y \in B) \wedge (y \in B \vee y \in A)$$

$$\text{as } y \in B, \text{ so } y \in A \therefore p \vee q, \neg p \therefore q \text{ Disjunctive syllogism.}$$

$$\text{so } y \in B \cap C \rightarrow y \in A \cap C$$

$$B \cap C \subseteq A \cap C \quad \textcircled{2}$$

$$\textcircled{1} \text{ \& } \textcircled{2} \rightarrow A \cap C = B \cap C$$

$$2) A \cap B \subseteq C$$

iff

$$A \cup C = B \cup C$$

$$\forall x [x \in A \cap B \rightarrow x \in C] \rightarrow \forall y [y \in A \cup C \rightarrow y \in B \cup C]$$

$$\forall x [(x \in A \wedge x \in B) \vee (x \in B \wedge x \in A)] \rightarrow x \in C \quad \textcircled{1} \quad \forall y [y \in B \cup C \rightarrow y \in A \cup C]$$

$$\text{suppose } A \cup C \neq B \cup C$$

$$\text{then } \exists y [y \in A \cup C \wedge y \notin B \cup C]$$

$$\forall y [y \in A \vee y \in C \rightarrow y \in B \vee y \in C]$$

$$\forall y [y \in B \vee y \in C \rightarrow y \in A \vee y \in C]$$

$$\exists y [(y \in A \vee y \in C) \wedge (y \notin B \wedge y \notin C)] \quad \textcircled{2}$$

then

$$\exists y [(y \in A \cup C \wedge y \notin B \cup C) \vee (y \in B \cup C \wedge y \notin A \cup C)]$$

$$\exists y [(y \in A \vee y \in C) \wedge (y \notin B \wedge y \notin C)] \quad \textcircled{2}$$

$$\exists y [(y \in B \vee y \in C) \wedge (y \notin A \wedge y \notin C)] \quad \textcircled{3}$$

Case 1) suppose ② $y \notin B \wedge y \notin C$. As $y \in A \vee y \in C$, so $y \in A$ since $y \notin C$. But as $y \in A \wedge y \notin B$ so from ① $\rightarrow y \notin C$ which is a contradiction or then $A \supset B \not\subseteq C$.

Case 2) suppose ③ $y \notin A \cdot y \notin C$. Since $y \in B \vee y \in C \& y \notin C$. So $y \in B$. But from ①, $y \in C$ which is again a contradiction or simply $A \supset B \not\subseteq C$ as $y \in B \wedge A \& y \notin C$.



$$A \cup C = B \cup C$$

$$A \supset B \subseteq C$$

$$\forall y [(y \in A \vee y \in C) \rightarrow (y \in B \vee y \in C)] \quad \text{①} \quad \forall x [(x \in A \wedge x \notin B) \rightarrow (x \in B \vee x \notin A)] \rightarrow x \in C$$

$$\forall y [(y \in B \vee y \in C) \rightarrow (y \in A \vee y \in C)] \quad \text{②}$$

suppose $A \supset B \not\subseteq C$.

$$\exists x [(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A) \wedge x \notin C] \quad \text{③}$$

$x_0 \notin C$

Case 1) $x_0 \in A \wedge x_0 \notin B$: since $x_0 \in A \& \text{so } x_0 \notin B \vee x_0 \in C$.

But as $x_0 \notin C$, so $x_0 \in B$. Now we have that $x \in$ contradiction or if $x \notin B$ then $x_0 \in C$. Again a contradiction.

Case 2) $x_0 \in B \wedge x_0 \notin A$. So $x_0 \in A \vee x_0 \in C$. Since $x_0 \notin A$ so $x_0 \in C$ which is a contradiction.

Remember $\sim(p \text{ iff } q)$

$$= \sim[(p \rightarrow q) \wedge (q \rightarrow p)]$$

$$= \sim[(\sim p \vee q) \wedge (\sim q \vee p)]$$

$$= \sim[(p \wedge \sim q) \vee (q \wedge \sim p)]$$

$$22) C \subseteq A^c B \text{ iff } (C \subseteq A \cup B) \wedge (A \cap B \cap C = \emptyset)$$

→

$$\forall x [x \in C \rightarrow ((x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A))] \quad \text{--- (1)}$$

Suppose $y \in C$ is arbitrary (Good)

$$\exists z [z \in A \wedge z \in B \wedge z \in C]$$

$$\text{Then (1)} \rightarrow (y \in A \wedge y \notin B) \vee (y \in B \wedge y \notin A) \rightarrow \forall z [z \notin A \vee z \notin B \vee z \notin C]$$

Case 1) $y \in A \wedge y \notin B$, then $y \in A$.

$$\rightarrow \forall z [z \in C \rightarrow (z \notin A \vee z \notin B)]$$

Case 2) $y \in B \wedge y \notin A$, then $y \in B$.

$$\therefore \sim p \vee q = p \rightarrow q$$

Suppose $z \in C$ is arbitrary

$$z \notin A \vee z \notin B$$

$$\text{then from (1)} (z \in A \wedge z \notin B) \vee (z \in B \wedge z \notin A)$$

Case 1) $z \in A \wedge z \notin B$ then $z \notin B$

Case 2) $z \in B \wedge z \notin A$ then $z \notin A$

←

$$\text{Supp } C \subseteq A \cup B \wedge A \cap B \cap C = \emptyset \rightarrow C \subseteq A^c B$$

$$\forall y [y \in C \rightarrow (y \in A \vee y \in B)] \quad \text{--- (1)}$$

$$\forall x [x \in C \rightarrow ((x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A))]$$

$$\forall z [z \notin A \vee z \notin B \vee z \notin C]$$

$$\forall z [z \in C \rightarrow (z \notin A \vee z \notin B)] \quad \text{--- (2)}$$

Let x be arbitrary

$$(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)$$

Supp $x \in C$.

$$\text{--- (3)} \rightarrow x \in A \vee x \in B$$

$$\text{--- (2)} \rightarrow x \notin A \vee x \notin B$$

Consider case 1 of (3)

$x \in A$ ~~$x \in B$~~ . Then from (2) $x \notin B$.

Case 2) $x \in B$. Then from (2) $x \notin A$.



$$23) a) A \setminus C \subseteq (A \setminus B) \cup (B \setminus C)$$

$$\forall x [x \in A \setminus C \rightarrow x \in (A \setminus B) \cup (B \setminus C)]$$

$$\text{Supp } x \in A \setminus C$$

$$(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin C)$$

$$x \in A \wedge x \notin C$$

Proof By Contradiction

$$\text{Suppose } A \setminus C \not\subseteq (A \setminus B) \cup (B \setminus C)$$

$$\text{then } \exists x [x \in A \setminus C \wedge x \notin (A \setminus B) \cup (B \setminus C)]$$

$$x \in A \wedge x \notin C$$

$$\neg [(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin C)]$$

$$(x \notin A \vee x \in B) \wedge (x \notin B \vee x \in C) \text{ --- (1)}$$

Since $x \in A$, so $x \in B$ from (1) $\therefore$ first given

& since $x \notin C$, so $x \notin B$ from (1). But this is a contradiction

$$(b) A^3 C \subseteq (A^3 B) \cup (B^3 C)$$

$$\text{Suppose } A^3 C \not\subseteq (A^3 B) \cup (B^3 C)$$

$$\text{then } \exists x [x \in A^3 C \wedge x \notin (A^3 B) \cup (B^3 C)] \text{ --- (1)}$$

$$(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A)$$

$$\neg [(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)] \vee [(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)]$$

$$[(x \notin A \vee x \in B) \wedge (x \notin B \vee x \in A)] \wedge [(x \notin B \vee x \in C) \wedge (x \notin C \vee x \in B)] \text{ --- (2)}$$

There are 4 given

Case 1) $x \in A \wedge x \notin C$. Since $x \in A$, so $x \in B$ $\therefore$ (2) But here since $x \in B$, so $x \in C$ which is a contradiction.

Case 2) $x \in C \wedge x \notin A$. Since $x \in C$, so $x \in B$. But then as $x \in B$, so $x \in A$ which is a contradiction so both cases lead to contradiction under assumption that $A \cap B \neq \dots$ so $A \cap B$ must be subset.

$$24) (A \cup B)^3 C \subseteq (A^3 C) \cup (B^3 C)$$

$$\forall x [x \in (A \cup B)^3 C \rightarrow x \in (A^3 C) \cup (B^3 C)]$$

$$\text{Supp}(x \in A \cup B \wedge x \notin C) \cap (x \in C \wedge x \notin A \cup B)$$

$$[(x \in A \vee x \in B) \wedge x \notin C] \vee [x \in C \wedge x \notin A \wedge x \notin B] \quad - (1)$$

We have to prove that

$$[(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A)] \vee [(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)]$$

① $\rightarrow$ Case 1)

$(x \in A \vee x \in B) \wedge x \notin C$. If $x \in A \wedge x \notin C$ then goal follows

if $x \in B \wedge x \notin C$. then goal follows.

Case 2) $x \in C \wedge x \notin A \wedge x \notin B$. Since $x \in C \wedge x \notin A$. Goal follows

Since $x \in C \wedge x \notin B$. Goal follows.

So each case lead to one of 4 cases in goal.

$$(b) A = \{1, 2, 3\} \quad B = \{2, 4, 5\} \quad C = \{1, 4, 6\}$$

Notice we kept $y = 4$ in $B \cap C$ but not in A .

$$25) (A^3 C) \cap (B^3 C) \subseteq (A \cap B)^3 C$$

Suppose $(A^3 C) \cap (B^3 C) \not\subseteq (A \cap B)^3 C$. Then there exist x such that

$$[(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A)] \wedge [(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)]$$

There are 2 given

$$(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A) \quad \text{--- (1)}$$

$$(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B) \quad \text{--- (2)}$$

~~E.H.S~~

$$\& \neg [(x \in A \wedge x \in B \wedge x \notin C) \vee (x \in C \wedge (x \notin A \vee x \notin B))]$$

$$= (x \notin A \vee x \notin B \vee x \in C) \wedge [x \notin C \vee (x \in A \wedge x \in B)]$$

$$= \text{--- (1)} \quad \wedge [(x \notin C \vee x \in A) \wedge (x \notin C \vee x \in B)]$$

There are 3 give

$$(x \notin A \vee x \notin B \vee x \in C) \quad \text{--- (3)}$$

$$x \notin C \vee x \in A \quad \text{--- (4)}$$

$$x \notin C \vee x \in B \quad \text{--- (5)}$$

~~From (1) Case 1) $x \in A \wedge x \notin C$. Then from (3) $x \notin B$. Then~~

~~from (5) $x \notin C$. But~~

We will prove by direct approach
& using contradiction in it.

$$\text{Suppose } (A^3 C) \cap (B^3 C) \subseteq (A \cap B)^3 C$$

Our givens are

$$(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A) \quad \text{--- (1)}$$

$$(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B) \quad \text{--- (2)}$$

We have to prove that

$$(x \in A \wedge x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin A) \vee (x \in C \wedge x \notin B) \quad (3)$$

Case 1) $x \in A \wedge x \notin C$ ($\therefore$ from 0). Suppose $x \notin B$.

Then from 2) $x \in C \wedge x \notin B$ but this is a contradiction so $x \in B$. The $x \in A \wedge x \in B \wedge x \notin C$ proves 1st case of goal.

Case 2) $x \in C \wedge x \notin A$. This proves 2nd case of goal directly.

Case 3) $x \in B \wedge x \notin C$. Suppose $x \notin A$. Then from 0) $x \in C$ $\wedge x \notin A$ but this is a contradiction over $x \notin C$. So $x \in A$.

This again proves First case of goal.

Case 4) $x \in C \wedge x \notin B$. This proves 3rd case of goal.

(b) One counter example is $A = \{2, 3\}$, $B = \{1, 4\}$, $C = \{1, 5\}$.

Note that $(A \cap B)^c \subseteq (A^c \cap B^c) \cup (A^c \cap C^c) \cup (B^c \cap C^c)$ means

$$(x \in A \wedge x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin A) \vee (x \in C \wedge x \notin B) \text{ Given}$$

$$\text{Goal: } \underbrace{(x \in A \wedge x \notin C)}_{(1)} \vee \underbrace{(x \in C \wedge x \notin A)}_{(2)} \wedge \underbrace{[(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)]}_{(4)}$$

Assuming $x \in A \wedge x \in B \wedge x \notin C$ proves 2 goals.

① & ⑤. One case should prove both goals at once as their is conjunction.

Now assume $x \in C \wedge x \notin A$ proves only ② which proves 1st part of goal not second but their is conjunction on both of them must be satisfied at once.

This also helped us to find counter example by taking $1 \in B \wedge 1 \in C$ but $1 \notin A$.

$$32) A \subseteq B \cup C$$

$$A \subseteq B \vee A \cap C \neq \emptyset$$

$$\forall x [x \in A \rightarrow (x \in B \vee x \in C)]$$

$$\forall y [y \in A \rightarrow y \in B] \vee$$

$$\exists x [x \in A]$$

$$\exists x [x \in A \wedge x \in C]$$

$$\text{Hw } y \in B \vee y \in C$$

Case 1) if $y \in B$ Hw $A \subseteq B$

Case 2) if $y \in C$ Hw $\text{sim } y \in A \cdot A \cap C \neq \emptyset$

The main difference in $\exists \& \forall$ quantifier is $\forall \& \exists$.

In 31. There can be case when both $A \subseteq B \wedge A \subseteq C$ are not satisfied if for some $y \in A$, $y \in B$ & for other $y \notin A$ & $y \notin B \& y \in C$.

But in 32) if we have all $y \in A \rightarrow y \in B$ Hw $A \subseteq B$. If even one $y \notin B$ Hw $y \in C$ & $A \cap C \neq \emptyset$.

33)

$$\exists x (P(x) \rightarrow \forall y P(y))$$

$$\exists x (\neg P(x) \vee \forall y P(y))$$

Case 1) Suppose $\forall y P(y)$. Since the domain is nonempty so $\forall y P(y)$ implies that some $P(a)$ is also true. Since all $P(y)$ are T, so $P(a) \rightarrow \forall y P(y)$. ($\because T \rightarrow T = T$) so certainly $\exists x (P(x) \rightarrow \forall y P(y))$

Case 2) Suppose $\neg \forall y P(y) = \exists y \neg P(y)$ let that y be a so $P(a)$ is F. So $P(a) \rightarrow \forall y P(y)$ ($\because F \rightarrow F = T$) so certainly $\exists x (P(x) \rightarrow \forall y P(y))$

This is True for every possible domain because of Truth table of implication